

Am25LS2539

Dual One-of-Four Decoder with Three-State Outputs and Polarity Control

DISTINCTIVE CHARACTERISTICS

- Two independent decoders/demultiplexers
- Three-state outputs
- Buffered common polarity control
- A. C. parameters specified over operating temperature and power supply ranges

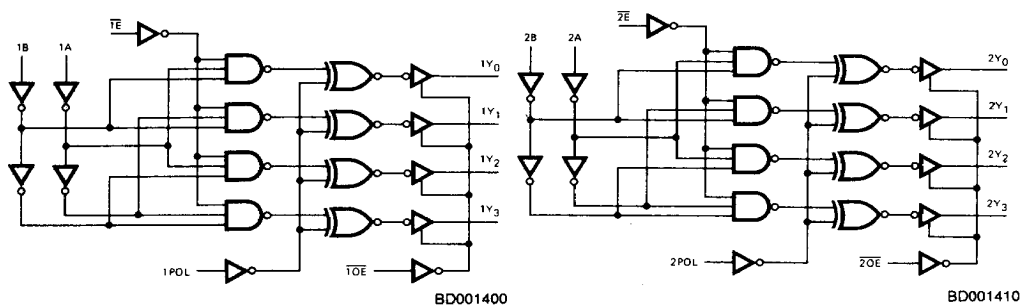
GENERAL DESCRIPTION

The Am25LS2539 is a dual two-line to four-line decoder/demultiplexer fabricated using advanced Low-Power Schottky technology. Each decoder has two buffered select inputs – A and B which are decoded to one-of-four Y outputs. An enable input ($\bar{E}$) is used for gating or can be used as a data input for demultiplexing applications. When the enable input goes HIGH, all four decoder functions are inhibited.

An output enable ($\bar{OE}$) input is used to control the three-state outputs of the device. When the $\bar{OE}$ input is LOW, the outputs are enabled. When the $\bar{OE}$ input is HIGH, the

outputs are in the high-impedance (off) state. The device also has separate buffered polarity (POL) inputs to force the outputs to either an active-HIGH state or an active-LOW state. When the POL input is LOW, the outputs are active-HIGH and when the POL input is HIGH, the outputs are active-LOW. The device is packaged in a space saving (0.3-inch row spacing) 20-pin package. The device features Am25LS family improved switching specification, higher noise margin, and twice the fan-out over the military temperature range when compared with Am54LS/74LS devices.

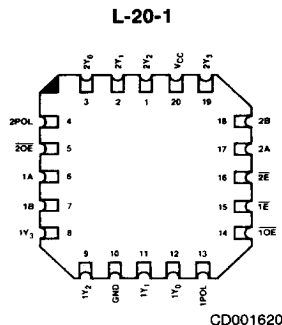
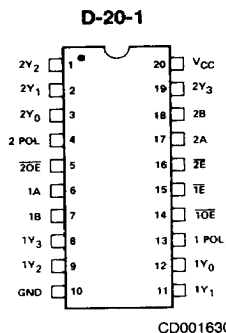
BLOCK DIAGRAM



RELATED PRODUCTS

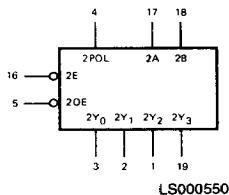
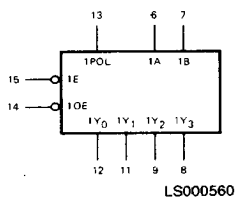
Part No.	Description
Am25LS2536	8-Bit Decoder
Am25LS2537	1-of-10 Decoder
Am25LS2538	1-of-8 Decoder
Am25LS2548	Chip Select Address Decoder
Am2921	1-of-8 Decoder
Am2924	3-to-8 Line Decoder/Demultiplexer

CONNECTION DIAGRAM Top View

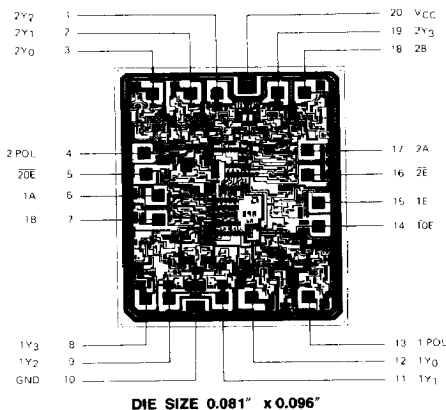


Note: Pin 1 is marked for orientation

LOGIC SYMBOL



METALLIZATION AND PAD LAYOUT



ORDERING INFORMATION

AMD products are available in several packages and operating ranges. The order number is formed by a combination of the following: Device number, speed option (if applicable), package type, operating range and screening option (if desired).

Am25LS2539

D

C

B

Screening Option
Blank - Standard processing
B - Burn-in

Temperature (See Operating Range)
C - Commercial (0°C to +70°C)
M - Military (-55°C to +125°C)

Package
D - 20-pin Cerdip
F - 20-pin flatpak
L - 20-pin leadless chip carrier
P - 20-pin plastic DIP
X - Dice

Device type
Dual 1-of-4 Decoder

Valid Combinations

Am25LS2539	PC
	DC, DM
	FM
	LC, LM
	XC, XM

Valid Combinations

Consult the AMD sales office in your area to determine if a device is currently available in the combination you wish.

PIN DESCRIPTION

Pin No.	Name	I/O	Description
	A, B	I	Select the two select inputs to the decoder/demultiplexer.
15, 16	$\bar{E}$ Enable	I	The enable input to the decoder. A HIGH input forces the decoding functions to be inhibited regardless of the A and B inputs.
13, 4	POL	I	Polarity Input. The polarity input forces the outputs to either an active-HIGH state or an active-LOW state. A LOW on the polarity input forces the output active-HIGH. A HIGH on the polarity input forces the outputs active-LOW.
14, 5	$\bar{OE}$	I	Output Enable. A LOW on the $\bar{OE}$ input enables the outputs. A HIGH on the $\bar{OE}$ inputs forces the outputs to the high high-impedance (off) state.
	Y_0, Y_1, Y_2, Y_3	O	The four decoder/demultiplexer outputs.

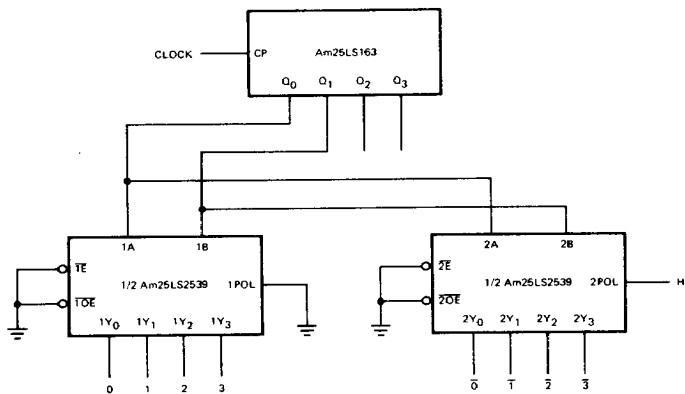
FUNCTION TABLE

Function	Inputs					Outputs			
	$\overline{OE}$	$\overline{E}$	POL	B	A	Y_0	Y_1	Y_2	Y_3
High-Impedance	H	X	X	X	X	Z	Z	Z	Z
Disable	L	H	L	X	X	L	L	L	L
	L	H	H	X	X	H	H	H	H
Active-High Output	L	L	L	L	L	H	L	L	L
	L	L	L	L	H	L	H	L	L
	L	L	L	H	L	L	L	H	L
	L	L	L	H	H	L	L	L	H
Active-Low Output	L	L	H	L	L	L	H	H	H
	L	L	H	L	H	H	L	H	H
	L	L	H	H	L	H	H	L	H
	L	L	H	H	H	H	H	H	L

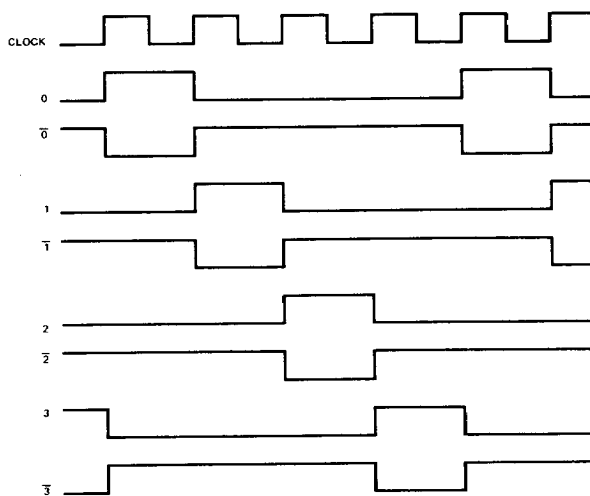
H = HIGH
L = LOW

X = Don't Care
Z = High-Impedance

APPLICATIONS



AF000831



AF000840

FOUR PHASE CLOCK GENERATOR

ABSOLUTE MAXIMUM RATINGS

Storage Temperature -65°C to +150°C
 Ambient Temperature Under Bias -55°C to +125°C
 Supply Voltage to Ground Potential
 Continuous -0.5V to +7.0V
 DC Voltage Applied to Outputs For
 High Output State -0.5V to +V_{CC} max
 DC Input Voltage -0.5V to +7.0V
 DC Output Current, Into Outputs 30mA
 DC Input Current -30mA to +5.0mA

Stresses above those listed under ABSOLUTE MAXIMUM RATINGS may cause permanent device failure. Functionality at or above these limits is not implied. Exposure to absolute maximum ratings for extended periods may affect device reliability.

OPERATING RANGES

Commercial (C) Devices

Temperature 0°C to +70°C
 Supply Voltage +4.75V to +5.25V

Military (M) Devices

Temperature -55°C to +125°C
 Supply Voltage +4.5V to +5.5V

Operating ranges define those limits over which the functionality of the device is guaranteed.

DC CHARACTERISTICS over operating range unless otherwise specified

Parameters	Description	Test Conditions (Note 2)		Min	Typ (Note 1)	Max	Units
V _{OH}	Output HIGH Voltage	V _{CC} = MIN V _{IN} = V _{IH} or V _{IL}	MIL, I _{OH} = -1.0mA COM'L, I _{OH} = -2.8mA	2.4 2.4	3.4 3.4		Volts
V _{OL}	Output LOW Voltage (Note 5)	V _{CC} = MIN V _{IN} = V _{IH} or V _{IL}	I _{OL} = 4.0 mA I _{OL} = 8.0mA I _{OL} = 12mA			0.4 0.45 0.5	Volts
V _{IH}	Input HIGH Level	Guaranteed input logical HIGH voltage for all inputs		2.0			Volts
V _{IL}	Input LOW Level	Guaranteed input logical LOW voltage for all inputs.				0.7 0.8	Volts
V _I	Input Clamp Voltage	V _{CC} = MIN, I _{IN} = -18mA				-1.5	Volts
I _{IL}	Input LOW Current	V _{CC} = MAX, V _{IN} = 0.4V				-0.36	mA
I _{IH}	Input HIGH Current	V _{CC} = MAX, V _{IN} = 2.7V				20	μA
I _I	Input HIGH Current	V _{CC} = MAX, V _{IN} = 7.0V				0.1	mA
I _{OZ}	Off-State (High-Impedance) Output Current	V _{CC} = MAX	V _O = 0.4V V _O = 2.4V			-20 20	μA
I _{SC}	Output Short Circuit Current (Note 3)	V _{CC} = MAX		-15		-85	mA
I _{CC}	Power Supply Current (Note 4)	V _{CC} = MAX			22	37	mA

- Notes: 1. Typical limits are at V_{CC} = 5.0V, 25°C ambient and maximum loading.
 2. For conditions shown as MIN or MAX, use the appropriate value specified under Operating Ranges for the applicable device type.
 3. Not more than one output should be shorted at a time. Duration of the short circuit test should not exceed one second.
 4. Test conditions: A = B = E = GND; POL = OE = 4.5V.
 5. V_{OL} is specified with total device I_{OL} = 60mA (max).

SWITCHING CHARACTERISTICS ($T_A = +25^\circ\text{C}$, $V_{CC} = 5.0\text{V}$)

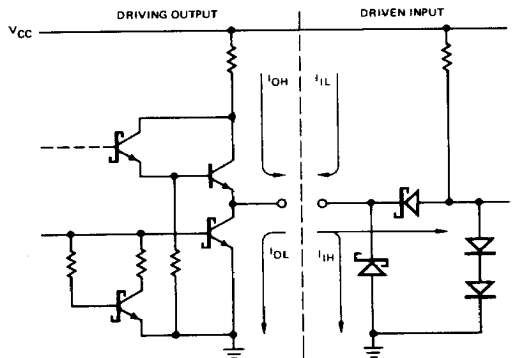
Parameters	Description	Test Conditions	Min	Typ	Max	Units
tPLH	A, B to Y _i	C _L = 15pF R _L = 2.0 kΩ		22	33	ns
tPHL				17	25	
tPLH	E to Y _i			19	28	ns
tPHL				21	31	
tPLH	POL to Y _i			16	24	ns
tPHL				19	28	
tZH	OE to Y _i			15	23	ns
tZL				15	22	
tHZ	OE to Y _i	C _L = 5.0 pF R _L = 2.0kΩ		19	28	ns
tLZ				23	34	

SWITCHING CHARACTERISTICS over operating range unless otherwise specified*

Parameters	Description	Test Conditions	COMMERCIAL		MILITARY		Units
			Am25LS2539		Am25LS2539		
			Min	Max	Min	Max	
tPLH	A, B, to Y _i	C _L = 50pF R _L = 2.0kΩ		41		48	ns
tPHL				34		42	
tPLH	E to Y _i			34		40	ns
tPHL				38		45	
tPLH	POL to Y _i			29		34	ns
tPHL				39		49	
tZH	OE to Y _i	C _L = 5.0pF R _L = 2.0kΩ		38		45	ns
tZL				24		25	
tHZ	OE to Y _i			33		37	ns
tLZ				36		37	

*AC performance over the operating temperature range is guaranteed by testing defined in Group A, Subgroup 9.

Am25LS2539 · Am54LS/74LS
LOW-POWER SCHOTTKY INPUT/OUTPUT
CURRENT INTERFACE CONDITIONS



Note: Actual current flow direction shown.